

Introduction to Node.js

1. What is Node.js?

-> Node.js ek JavaScript runtime hai jo browser ke bahar JS chalata hai. Fast hota hai kyunki V8 engine use karta hai.

Founder :- Ryan Dahl

Launch :- 2009

2. What is a Server?

-> Ek system hota hai jo client ko data ya service deta hai.

3. What is Cross-Platform?

-> Cross-platform ka matlab hota hai aisa software jo Windows, macOS aur Linux jaise multiple operating systems par bina dobara likhe chal sakta hai.

4. What is Server-Side Scripting?

-> Server-side scripting mein code server par run hota hai, browser par nahi. Isse database se data laakar user ko dynamic aur customized web pages milte hain.

5. Alternatives to Node.js :-

-> 1.PHP

2.Java

3.Ruby

4.Python

6. Uses of Node.js :-

-> 1.Real-time applications (e.g., chat applications, online gaming)

2.Data streaming

3.Wireless connectivity

4.System monitoring dashboards

5.Chatbots

6.RESTful APIs

7. What is REPL(Read-Eval-Print Loop)?

-> REPL ek interactive shell hai jo Node.js ke saath aata hai aur terminal me run hota hai. Isme aap JavaScript code line by line run kar sakte ho.

1.Read: User ka input leta hai

2.Eval: Code ko execute karta hai

3.Print: Result show karta hai

4.Loop: Ye process repeat hota rehta hai jab tak aap exit na karo (Ctrl+C ya Ctrl+D)

8. What is Global Objects in Node.js?

-> Node.js me kuch global objects hote hain jo bina import kiye kahin bhi use ho sakte hain.

1.__filename: Current file ka complete path batata hai

2.__dirname: Current file ke folder ka complete path batata hai

9. Node.js Modules types and Example.

-> Types of Modules :-

1.Core Modules

2.Local Modules

3.Third-Party Modules

1.Core Module :- Core modules Node.js me built-in hote hain, inhe install karne ki zarurat nahi hoti. Inhe require() se use karte hain.

Examples :-

- 1.http: HTTP server banane ke liye
- 2.fs: File system ke saath kaam karne ke liye
- 3.path: File aur folder paths handle karne ke liye
- 4.querystring: URL query string parse/format karne ke liye
- 5.url: URL ko parse aur handle karne ke liye

2.Local Modules :- Local modules user-defined hote hain jo Node.js app ke andar banaye jaate hain. Ye specific functionality ko alag files ya folders me organize karte hain.

3.Third-Party Modules :- Third-party modules npm se milte hain aur project me install kiye jaate hain.

Examples :-

- 1.mongoose: MongoDB ke saath kaam karne ke liye
- 2.multer: File upload handle karne ke liye
- 3.passport: Authentication ke liye
- 4.cookie-parser: Cookies parse karne ke liye

10. What is Express.js?

-> Express.js ek lightweight Node.js framework hai jo server banana aur HTTP request/response handle karna easy banata hai.

11. MVC Framework :-

- > 1.Model(M) :- Data aur logic handle karta hai.
2.View(V) :- User ko data show karta hai (UI).
3.Controller(C) :- User input handle karta hai.
4.EJS (Embedded JavaScript) :- EJS ek templating engine hai jo JavaScript se dynamic HTML banata hai. Express ke saath use hota hai.

12. Installation of Express.js and EJS :-

- > 1.npm install express
2.npm install ejs

13. What is Nodemon?

-> File change hote hi server auto-restart karta hai.

```
"scripts": {  
  "start": "nodemon index.js"  
}
```

Note :- EJS files ko views folder me .ejs extension ke saath rakho

14. What is a Template engine?

-> Template engine Node.js me static HTML files me dynamic data inject karne ke kaam aata hai, jisse browser ko final HTML milta hai.

Example: EJS, Pug, Handlebars, Nunjucks

15. What is a Middleware?

-> Middleware Node.js ka ek function hota hai jo request aur response ke beech kaam karta hai. Isme req, res aur next() hota hai, jisse control next middleware ko pass hota hai.

16. What is a Database?

-> Database ek organized electronic data collection hota hai jisme data ko store, manage aur retrieve kiya jata hai.

Database ke Two(02) types:-

1.Relational Database (SQL)

Example: MySQL, PostgreSQL, Oracle

2.Non-Relational Database (NoSQL)

Example: MongoDB

Other Databases:-

1.Hierarchical

2.Cloud

3.Object-Oriented Database (OOD)

17. What is Schema?

-> Schema ek blueprint hota hai jo database me data ka structure define karta hai.

18. Write Mongoose Uses :-

-> 1.Database Connection Management – Node.js ko MongoDB se easily connect karta hai.

2.Schema Definition – Data ka structure define karta hai.

3.CRUD Operations – Create, Read, Update, Delete easily karta hai.

4.Data Validation – Galat data save hone se rokta hai.

5.Middleware (Hooks) – Save ya update se pehle/baad code run karne deta hai.

